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| Please fill in ythe manuscript title.                                       | <b>Well Kick Detection and Quantification Using Real-Time Anomaly Detection</b> |             |
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## Abstract

*Well control remains one of the critical challenges in present-day drilling operations, where the inadvertent influx of formation fluids—known as a kick—can rapidly escalate into a blowout if not detected and mitigated immediately. This paper presents a comprehensive approach to well kick detection, leveraging state-of-the-art anomaly detection techniques integrated with advanced artificial intelligence (AI) and machine learning (ML) models. By employing real-time data streams from drilling operations, the proposed methodology fuses deep learning architectures with classical well control calculations to predict kick events and quantify their volumes. The approach is grounded by rigorous mathematical formulations, data cleaning and feature engineering pipelines, and ensemble learning techniques that integrate the strengths of multiple predictive models. The resulting system is capable of providing prompt alerts and accurate estimations of kick volume, which are critical for proactive well control and prevention of catastrophic failures.*

**Keywords:** Well Kick Detection, Real-Time Anomaly Detection, Deep Learning, Ensemble Learning, Drilling Operations, Well Control, Machine Learning, Data Cleaning, Feature Engineering.

## 1. Introduction

### 1.1 Background and Motivation

Modern drilling operations have evolved to incorporate advanced sensors and data acquisition systems, generating vast quantities of real-time data. These data include measurements of mud flow rates, pressures, temperatures, and depths, which collectively offer a detailed view of the well’s dynamic behavior. The integration of artificial intelligence and machine learning techniques into this domain has opened new avenues for anomaly detection and predictive analytics, enabling operators to identify abnormal patterns indicative of kick events.

## 1.2 Problem Statement

A well kick occurs when the formation pressure exceeds the hydrostatic pressure of the drilling mud, causing an influx of formation fluids into the wellbore. Mathematically, a kick can be characterized by the condition:

$$P_f > P_h$$

where  $P_f$  is the formation pressure and  $P_h$  is the hydrostatic pressure of the mud column. The primary challenge lies not only in detecting the occurrence of a kick but also in quantifying its volume accurately. A small influx might be manageable, while a large influx—if not controlled promptly—can escalate into a catastrophic blowout.

## 1.3 Objectives of the Study

- **Develop a Real-Time Kick Detection System:** Design and implement a machine learning framework that leverages real-time data to detect kicks using anomaly detection techniques.
- **Integrate Physics-Based Volume Estimation:** Combine the ML predictions with classical well control equations to accurately estimate kick volumes.
- **Employ an Ensemble Learning Approach:** Utilize an ensemble meta-learner that fuses predictions from multiple base models (e.g., Random Forest, LSTM, and PINN) to improve robustness and accuracy.
- **Provide a User-Centric Solution:** Develop a system that not only predicts kick events and their volumes but also presents the results in an actionable format for field operators.

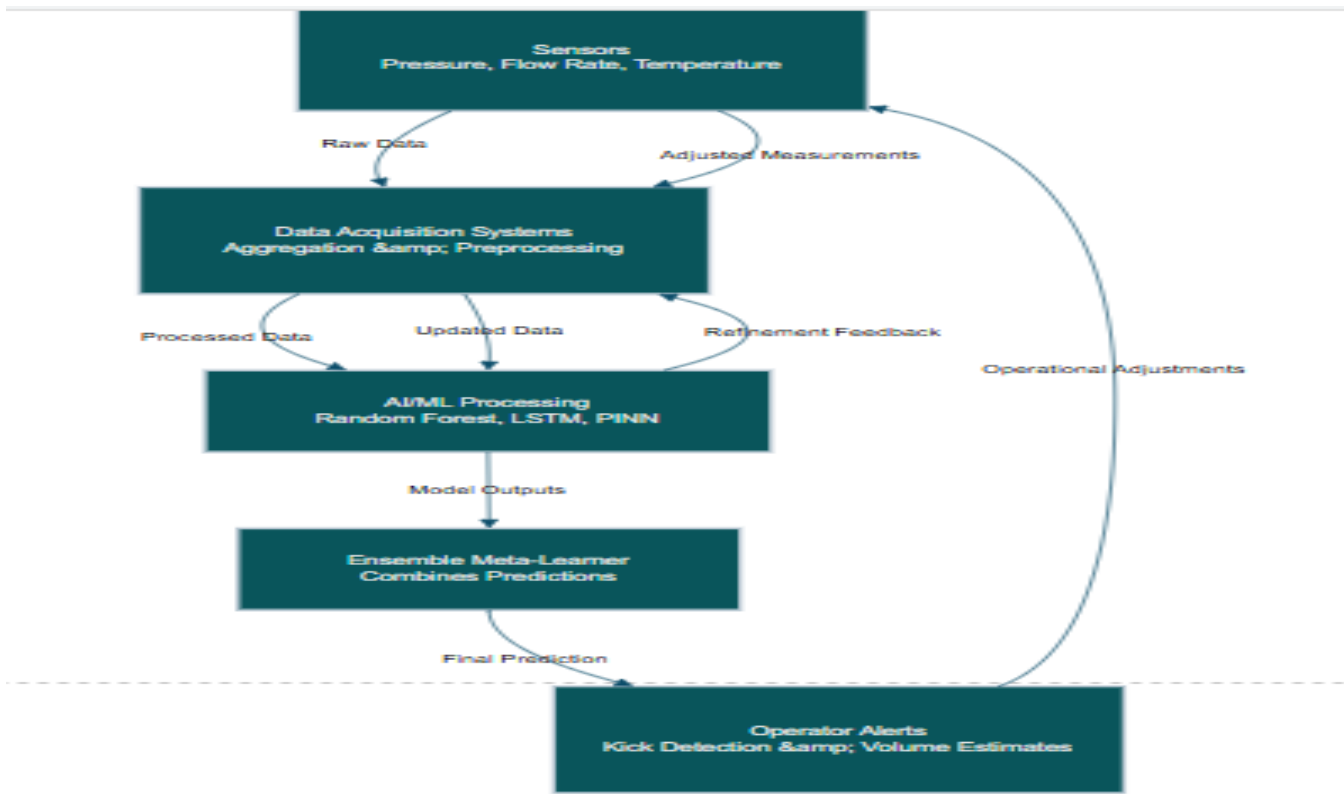


Fig 1.1 Overview of Data Flow in Modern Drilling Operations

## 1.4 Scope and Significance

This paper is structured to provide a comprehensive view of the development and integration of the well kick detection system. It covers the data acquisition and cleaning processes, feature engineering techniques, model selection and training, and the integration of classical engineering calculations with modern AI methods. The significance of this work lies in its potential to transform well control practices by providing a proactive and reliable method for kick detection and volume estimation, ultimately enhancing safety and reducing operational costs.

## 2. Literature Review

### 2.1 Traditional Methods for Kick Detection

Historically, well kick detection has relied on direct measurements and monitoring of drilling parameters. Conventional techniques include:

- **Mudlogging and Gas Analysis:** Mudlogging involves continuous sampling of the drilling mud for hydrocarbon gases and cuttings. Indicators such as sudden increases in gas concentration or changes in drilling cuttings are used to infer the presence of a kick. Although effective, these methods are inherently delayed due to the time required for samples to be transported from the wellbore to the surface.
- **Pit Gain Monitoring:** The concept of pit gain monitoring is based on the principle that, during a kick, the volume of fluid in the mud pits increases unexpectedly. Mathematically, the net volume of influx can be expressed as:

$$V_{kick} = V_{pumped} - V_{returned}$$

where  $V_{pumped}$  is the volume of mud pumped into the well and  $V_{returned}$  is the volume measured in the return line. However, this method is sensitive to errors arising from measurement inaccuracies and fluid losses.

- **Flow Rate and Pressure Monitoring:** A critical parameter in well control is the balance between formation pressure and hydrostatic pressure. The condition for a kick is often defined by:

$$P_f > P_h$$

where  $P_f$  represents the formation pressure and  $P_h$  is the hydrostatic pressure exerted by the drilling fluid. Hydrostatic pressure itself is calculated using:

$$P_h = TVD \times Mud\ Weight \times 0.052$$

In this equation, TVD denotes the true vertical depth in feet, and the factor 0.052 converts the product into pounds per square inch (psi). Traditional well control procedures focus on maintaining the mud weight such that the hydrostatic pressure counteracts formation pressures, thereby minimizing the risk of influx.

- **Annular Volume Estimation:** When a kick occurs, the volume of the influx is influenced not only by the net flow difference but also by the geometry of the wellbore. A widely used formula to estimate kick volume is:

$$V_{kick} = (V_{pumped} - V_{returned}) + (A_{annulus} \times \Delta L)$$

Here,  $A_{annulus}$  is the annular area, which can be computed as:

$$A_{annulus} = 4\pi(D_{hole}^2 - D_{pipe}^2)$$

and  $\Delta L$  represents the vertical distance (in feet) over which the influx occurs. This approach, although grounded in fundamental principles of fluid mechanics, requires precise measurements and is subject to uncertainties in wellbore geometry and fluid properties.

### 2.2 Advances in Real-Time Sensor-Based and Data-Driven Approaches

With the advent of high-resolution sensors and automated data acquisition systems, the drilling industry has witnessed a paradigm shift toward real-time monitoring and analysis. Key advancements include:

- **Digital Mudlogging Systems:** Modern mudlogging systems integrate real-time gas detection and sensor networks that monitor parameters such as mud density, flow rates, and temperature. These systems utilize data analytics to provide early warnings of abnormal conditions that may indicate a kick.
- **Pit Volume and Flow Analysis with Real-Time Data:** Improvements in sensor technology have enhanced the accuracy of pit gain measurements. Real-time data processing algorithms continuously evaluate changes in pit volumes and flow rates. Anomaly detection algorithms flag deviations from expected patterns, allowing for prompt intervention.

- **Integration of Distributed Acoustic Sensing (DAS):** DAS technology uses fiber-optic cables to measure acoustic signals along the wellbore. Variations in the acoustic signatures can indicate fluid influx and provide spatially continuous data about the well's behavior. This method offers the potential for earlier detection of kicks compared to traditional discrete sensors.
- **Application of Statistical Process Control (SPC):** Statistical methods have been employed to monitor the stability of drilling operations. Techniques such as control charts and cumulative sum (CUSUM) methods are used to detect small shifts in drilling parameters that precede kick events.

### 2.3 Deep Learning and Machine Learning Applications in Drilling Operations

The evolution of machine learning (ML) and deep learning (DL) techniques has opened new frontiers in anomaly detection, particularly for complex and non-stationary environments like drilling operations. Recent research has focused on the following approaches:

**Long Short-Term Memory (LSTM) Networks:** LSTM networks are a variant of recurrent neural networks (RNNs) that excel at modeling time-series data. Their ability to capture long-term dependencies makes them ideal for detecting gradual trends that may lead to a kick. In the context, an LSTM network is employed to process sequential drilling data, learning patterns indicative of abnormal pressure conditions. The LSTM cell, characterized by its internal gates, computes its output as follows:

$$\begin{aligned}
 ft &= \sigma(Wf \cdot [ht - 1, xt] + bf) \\
 it &= \sigma(Wi \cdot [ht - 1, xt] + bi) \\
 C\sim t &= \tanh(WC \cdot [ht - 1, xt] + bC) \\
 Ct &= ft \odot Ct - 1 + it \odot C\sim t \\
 ot &= \sigma(Wo \cdot [ht - 1, xt] + bo) \\
 ht &= ot \odot \tanh(Ct)
 \end{aligned}$$

Here,  $xt$  is the input at time  $t$ ,  $ht - 1$  is the previous hidden state, and  $\sigma$  denotes the sigmoid activation function. These equations illustrate the mechanism by which LSTMs selectively remember and forget information, enabling robust detection of subtle anomalies.

**Ensemble Learning Approaches:** Ensemble learning is a powerful technique that combines the predictions of multiple machine learning models to improve overall performance. In this study, we employ an ensemble learning approach that integrates predictions from three distinct models: a Random Forest classifier, an LSTM network, and a Physics-Informed Neural Network (PINN). Each model captures different aspects of the drilling data, and their combined predictions are aggregated using a meta-learner. This stacking strategy enhances the robustness and accuracy of the system by leveraging the strengths of each individual model.

The meta-learner processes the concatenated probability outputs from the base models to produce a final prediction. This approach not only improves the reliability of kick detection but also provides a more nuanced understanding of the underlying patterns in the data. The ensemble prediction is mathematically represented as:

$$y^\wedge = f(\text{ensemble}([y^{RF}, y^{LSTM}, y^{PINN}]))$$

where  $y^{RF}$ ,  $y^{LSTM}$ , and  $y^{PINN}$  represent the probability predictions from the Random Forest, LSTM, and PINN models, respectively.

**Transformer-Based Models:** While LSTMs have been widely adopted for time-series forecasting, transformer architectures have recently shown promise in this domain. These models leverage self-attention mechanisms to weigh the importance of different time steps dynamically, offering an alternative for detecting temporal anomalies. However, their computational complexity and data requirements often make them less practical for certain drilling applications compared to LSTMs.

**Anomaly Detection Techniques:** Beyond supervised learning, unsupervised and semi-supervised methods such as autoencoders and isolation forests have been explored for anomaly detection. These techniques aim to learn a

representation of normal operating conditions and flag deviations as potential kick events. The reconstruction error in an autoencoder, for example, is given by:

$$E = \sum_{i=1}^n (x_i - \hat{x}_i)^2$$

where  $x$  is the input data and  $\hat{x}$  is its reconstruction. A high error  $E$  may indicate an anomaly.

## 2.4 Limitations and Challenges in Current Approaches

**Data Quality and Availability:** Drilling data are inherently noisy and often incomplete. Missing values, sensor drift, and variable sampling rates pose significant challenges for both classical and ML-based methods. Robust data cleaning and feature engineering are therefore critical prerequisites.

**Non-Stationarity of Drilling Data:** The dynamic nature of drilling operations means that the statistical properties of the data can change over time. Models must be adaptable and capable of handling non-stationary time-series data to remain effective in real-time applications.

**Integration of Physics-Based Models:** While data-driven models can capture complex patterns, integrating them with physical laws and engineering principles (such as fluid mechanics and wellbore geometry) is essential for interpretability and reliability. Hybrid approaches that combine ML predictions with classical equations for volume estimation have shown promise, but further research is needed to refine these methods.

**Computational Constraints:** Real-time kick detection requires fast, efficient algorithms that can operate on streaming data. High model complexity can lead to delays that undermine the benefits of early detection. Balancing model performance with computational efficiency remains an ongoing challenge.

**Interpretability:** Black-box models, particularly deep learning architectures, often lack transparency, making it difficult for field engineers to understand the basis for predictions. Enhancing interpretability through techniques such as SHAP (SHapley Additive exPlanations) and model-agnostic methods is an active area of research.

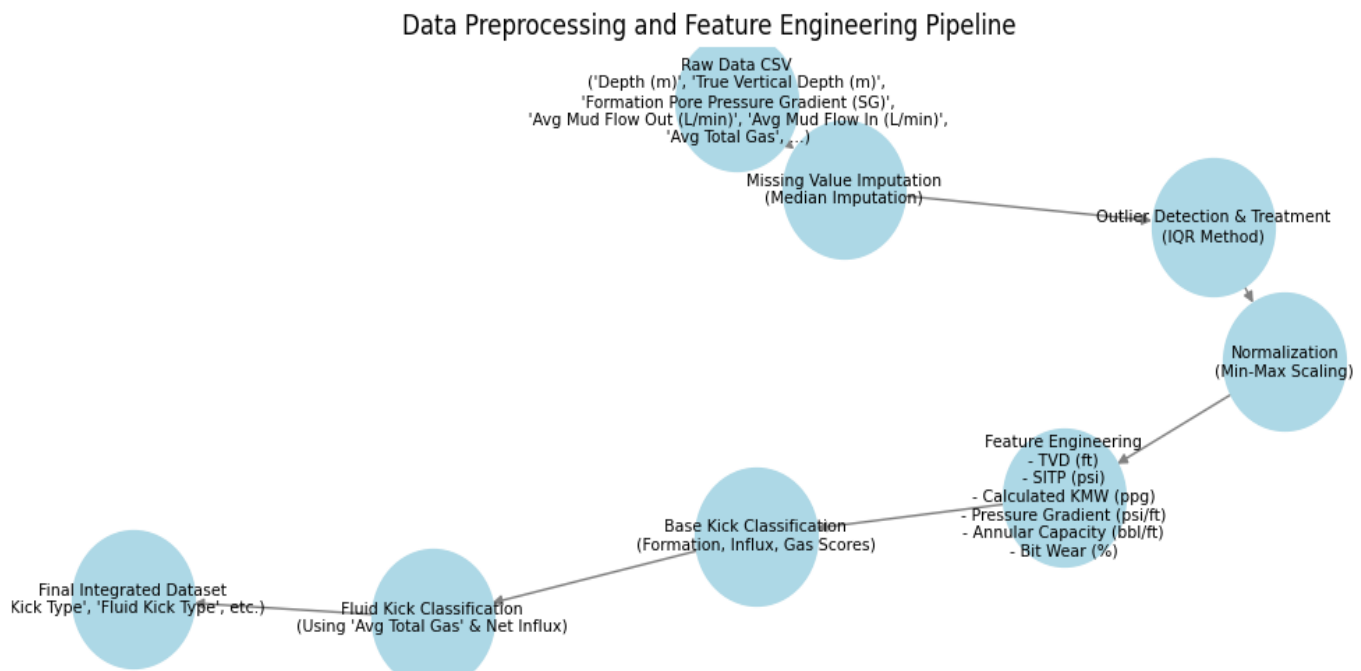


Fig 3.1 Data Preprocessing and Feature Engineering Pipeline

### 3. Data Preprocessing and Feature Engineering

#### 3.1 Overview and Rationale

Raw drilling datasets set of data from the Volve field, 2008-2016 released by Equinor and are typically recorded in CSV format and include a multitude of parameters such as "Formation Pore Pressure Gradient (SG)", "Avg Mud Flow Out (L/min)", "Avg Mud Flow In (L/min)", and "Avg Total Gas". These datasets often suffer from missing values, outliers, and disparate scales across features. The supporting document emphasizes key preprocessing objectives including data integrity, handling missing values via median imputation (especially in skewed distributions), outlier detection using methods like the interquartile range (IQR), normalization, and feature augmentation to capture critical physical phenomena.

- **Missing Value Imputation:** Missing data, common from sensor downtime, are imputed using the median of each feature  $X = \{x_1, x_2, \dots, x_n\}$
- **Outlier Detection and Treatment**  
Outliers, often from sensor spikes (e.g., in hookload or torque), are detected using the interquartile range (IQR):

$$IQR = Q3 - Q1 \quad \text{Range: } [Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$$

Values outside this range (e.g., extreme SITP or WOB) are clipped to boundaries or removed to stabilize model training.

#### Normalization

Features like mud flow  $\left(\frac{L}{min}\right)$  and pump pressure (Bar) vary widely in scale. Min-max normalization maps each to  $[-1, 1]$ :

$$x_{norm} = -1 + 2 \times \frac{(x - \min(x))}{(\max(x) - \min(x))}$$

This ensures equal weighting in predictive models, avoiding bias from high-magnitude parameters.

#### 3.2 Base Kick Classification and Derived Variable Computation

Central to the approach is the `classify_kick_dataset_spe` function, which implements a base kick classification scheme grounded in industry-standard SPE/IADC indicators. This function processes the input DataFrame by computing three primary indicators:

##### 1. Formation Pressure Anomaly:

Utilizing the "Formation Pore Pressure Gradient (SG)" parameter, a scoring function (`score_formation`) is applied to assess the pressure anomaly:

- **Score 0:** For values less than 1.05.
- **Score 2:** For values between 1.05 and 1.15.
- **Score 3:** For values between 1.15 and 1.25.
- **Score 4:** For values equal to or exceeding 1.25.

##### 2. Influx Volume Proxy:

The net mud flow is computed as the difference between "Avg Mud Flow Out (L/min)" and "Avg Mud Flow In (L/min)". This difference is then scored using the `score_influx` function:

- **Score 0:** When the difference is below 10 L/min.
- **Score 2:** When it falls between 10 and 20 L/min.
- **Score 3:** For differences between 20 and 30 L/min.
- **Score 4:** When the difference is 30 L/min or greater.

##### 3. Gas Indicator:

The "Avg Total Gas" reading is used to quantify the gas component during drilling. The scoring function (`score_gas`) assigns:

- **Score 0:** If the gas reading is below 150.

- **Score 2:** For readings between 150 and 250.
- **Score 3:** For readings between 250 and 400.
- **Score 4:** For values equal to or exceeding 400.

The individual scores from these three indicators are summed to yield a `total_score` (with a maximum of 12). A custom mapping function then translates this aggregate score into a kick classification—“No Kick”, “Minor Kick”, “Moderate Kick”, “Severe Kick”, “Blowout Risk”, or “Blow Out”—ensuring alignment with operational thresholds.

In addition to these base indicators, several derived variables are computed to enrich the dataset:

- **True Vertical Depth (TVD):** “True Vertical Depth (m)” is converted to feet by

$$TVD (ft) = TVD (m) \times 3.28084$$

- **SITP and Calculated KMW:** The “Avg Pump Pressure (Bar)” is converted to psi (SITP (psi)) as follows:

$$SITP (psi) = Avg Pump Pressure (Bar) \times 14.5038$$

- The Calculated Kick Mud Weight (KMW) in ppg is then computed by:

$$Calculated KMW (ppg) = \frac{SITP(psi)}{TVD (ft) \times 0.052} + orig\_mw$$

$$orig_{mw} = Original Mud Weight (ppg) \text{ otherwise, } Avg Mud Weight Out (SG) \times 8.34$$

- **Pressure Gradient:** The pressure gradient in *psi/ft* is calculated using:  

$$Pressure Gradient (psi/ft) = Avg Mud Weight Out (SG) \times 8.34 \times 0.052$$
- **Annular Capacity:** This is estimated from wellbore geometry by:  

$$Annular Capacity (bbl/ft) = 0.0009714 \times ((Casing Diameter (in))^2 - (Drill Pipe Diameter (in))^2)$$
- **Bit Wear:** Bit wear is assessed by comparing the current bit diameter to the initial measurement:  

$$Bit Wear (\%) = (Original Bit Diameter (in) - Bit Diameter (in)) / Original Bit Diameter (in) \times 100.$$

### 3.3 Fluid-Type Classification

The initial classification is further refined by the `classify_fluid_type` function. This function analyzes the fluid nature of a kick event by incorporating both the “Avg Total Gas” and the net mud flow difference. Specifically:

- If the base “Kick Type” is “No Kick”, the fluid classification remains “No Kick”.
- For gas readings equal to or exceeding 400, the event is subdivided into Minor, Moderate, or Severe Gas Kick based on finer thresholds. When gas levels are low (below 150) but the net influx is significant, the event is reclassified as a liquid kick, with further subdivisions based on the magnitude of the influx. In cases where neither condition is clearly met, the event is categorized as a “Mixed Kick”.

### 3.4 Classification Pipeline

The complete processing pipeline is orchestrated by the `integrated_classification` function. This function first invokes `classify_kick_dataset_spe` to generate base indicators and derived features. It then applies `classify_fluid_type` across the dataset to assign a “Fluid Kick Type” to each record. The final integrated dataset contains both the initial measurements and the comprehensive kick classifications, and is readily exportable to CSV format for further analysis or direct use in predictive modeling.

## 4. Experimental Setup and Results

### 4.1 Dataset Description

The dataset used in the experiments originates from high-frequency drilling data gathered during simulated drilling operations. The data comprise a comprehensive set of features including, but not limited to, "SITP (psi)", "Avg Pump Pressure (Bar)", "Formation Pore Pressure Gradient (SG)", "Avg Mud Flow In (L/min)", "Avg Mud Flow Out (L/min)", "Depth (m)", and "Annular Capacity (bbl/ft)" with a total of 12142 rows. These features capture both the operational and physical aspects of the drilling process. Prior to model training, the raw data underwent extensive cleaning and feature engineering as described in Section 3.

### 4.2 Data Splitting

- **Training Set:** 70% of the data, used to train the base models (Random Forest, LSTM, and PINN).
- **Validation Set:** 15% of the data, utilized for hyperparameter tuning and early stopping.
- **Test Set:** 15% of the data, reserved for final performance evaluation.

A stratified sampling approach was applied to preserve the relative frequency of kick events across the subsets. Furthermore, to facilitate time-series modeling, sequential samples were generated using a sliding window approach with a window size  $w=10$ . In this manner, for each time step  $i$  the sequence is defined as:

$$X(i) = \{x_i, x_{i+1}, \dots, x_{i+w-1}\}$$

### 4.3 Model Training and Hyperparameter Tuning

The training process involved optimizing three base models:

- **Random Forest Classifier:** Trained on tabular features using an ensemble of 100 decision trees. The model's hyperparameters (e.g., number of trees, maximum depth) were fine-tuned via grid search.
- **LSTM Network:** Designed to capture long-term temporal dependencies, the LSTM network comprises 50 hidden units and a dense layer with 20 neurons. Training was performed using the Adam optimizer with a learning rate of  $10^{-3}$  and early stopping based on validation loss.
- **Physics-Informed Neural Network (PINN):** This model integrates domain knowledge via a custom loss function. Its architecture mirrors that of the LSTM, with an additional physics-based regularization term in the loss function:

$$L_{total} = L_{classification} + \lambda L_{physics},$$

where  $\lambda$  is a weighting factor determined empirically. Subsequently, the outputs (i.e., probability vectors) from these base models were concatenated to form a **meta feature vector**:

$$z = [y^{RF}, y^{LSTM}, y^{PINN}],$$

which was input to an **ensemble meta-learner** that performed final classification using a stacking strategy. The meta-learner was trained on the validation set, optimizing for maximum overall accuracy.

### 4.4 Evaluation Metrics

- **Accuracy:** The proportion of correct predictions.
- **Precision, Recall, and F1-Score:** Evaluated for each class to gauge the model's performance in detecting kicks.
- **Confusion Matrix:** Provided insight into class-specific performance and misclassification patterns.

- **Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE):** Used to quantify the error in kick volume estimation.

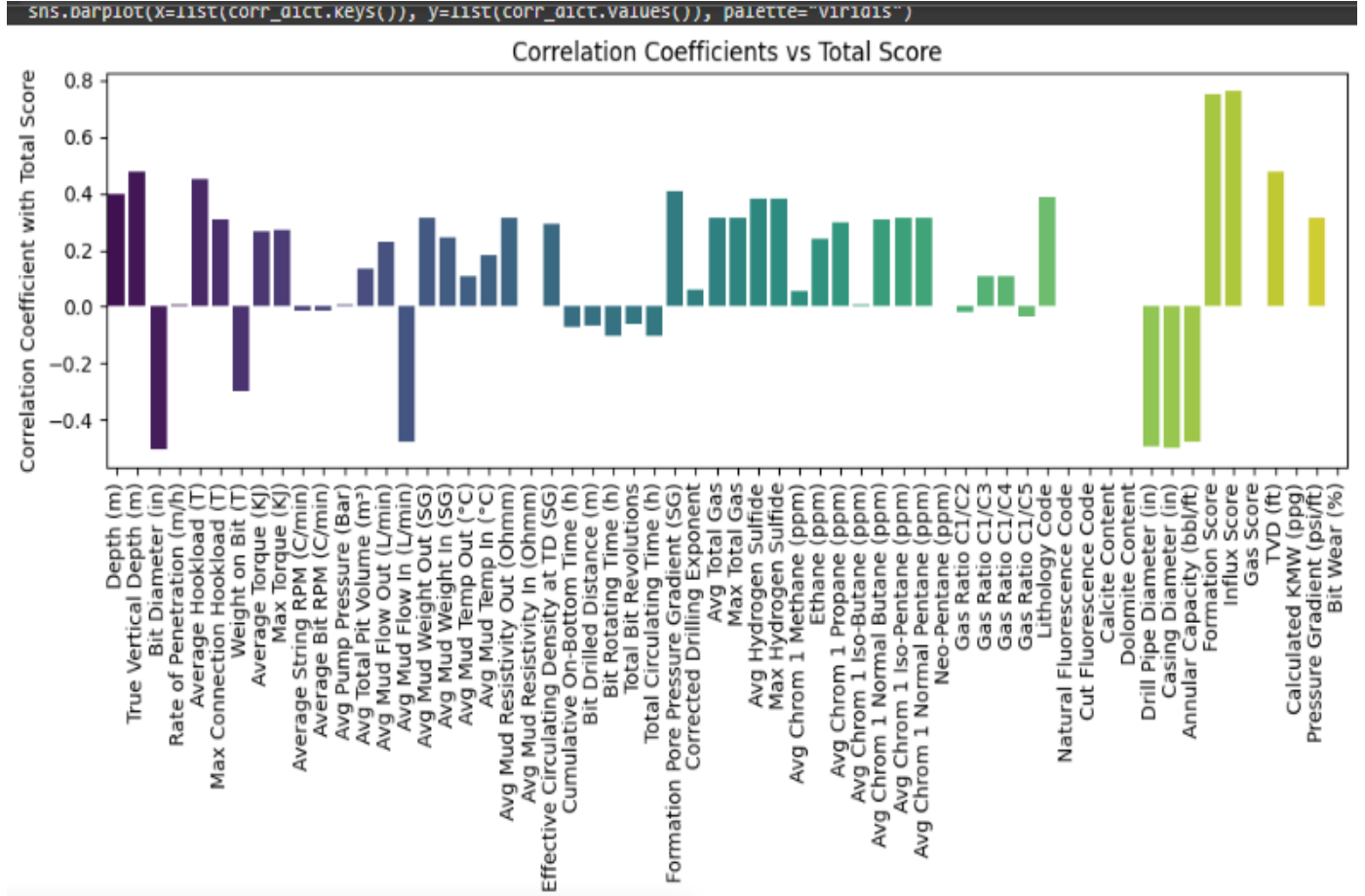


Fig4.1 Coefficients of correlation of each variables against the total score used to classify the kicks.

For the volume estimation, two components were computed:

- **Flow Difference Component (Vflow):**

$$V_{flow} = 158.987 \max(Q_{in} - Q_{out}, 0),$$

- **Annular Volume Component (Vannular):**

$$V_{annular} = A_{annulus} \times \Delta L,$$

where  $\Delta L$  is the change in depth (in feet) computed as:

$$\Delta L = Depth_{ft}(i) - Depth_{ft}(i - 1),$$

and the total estimated kick volume is given by:

$$V_{kick} = V_{flow} + V_{annular}.$$

## 4.5 Experimental Results

### 4.5.1 Classification Performance

Table 4.1: a table showing Classification Performance of the combined model to form an ensemble meta-learner

**Combined Model Test Accuracy: 0.9818930041152263**

| <i>Label</i>        | <b>Precision</b> | <b>Recall</b> | <b>F1-score</b> | <b>Support</b> |
|---------------------|------------------|---------------|-----------------|----------------|
| 0                   | 0.98             | 1.00          | 0.99            | 524            |
| 1                   | 0.96             | 0.96          | 0.96            | 163            |
| 2                   | 0.98             | 0.97          | 0.97            | 387            |
| 3                   | 0.99             | 0.99          | 0.99            | 141            |
| <b>Accuracy</b>     |                  |               | 0.98            | 1215           |
| <b>Macro avg</b>    | 0.98             | 0.98          | 0.98            | 1215           |
| <b>Weighted avg</b> | 0.98             | 0.98          | 0.98            | 1215           |

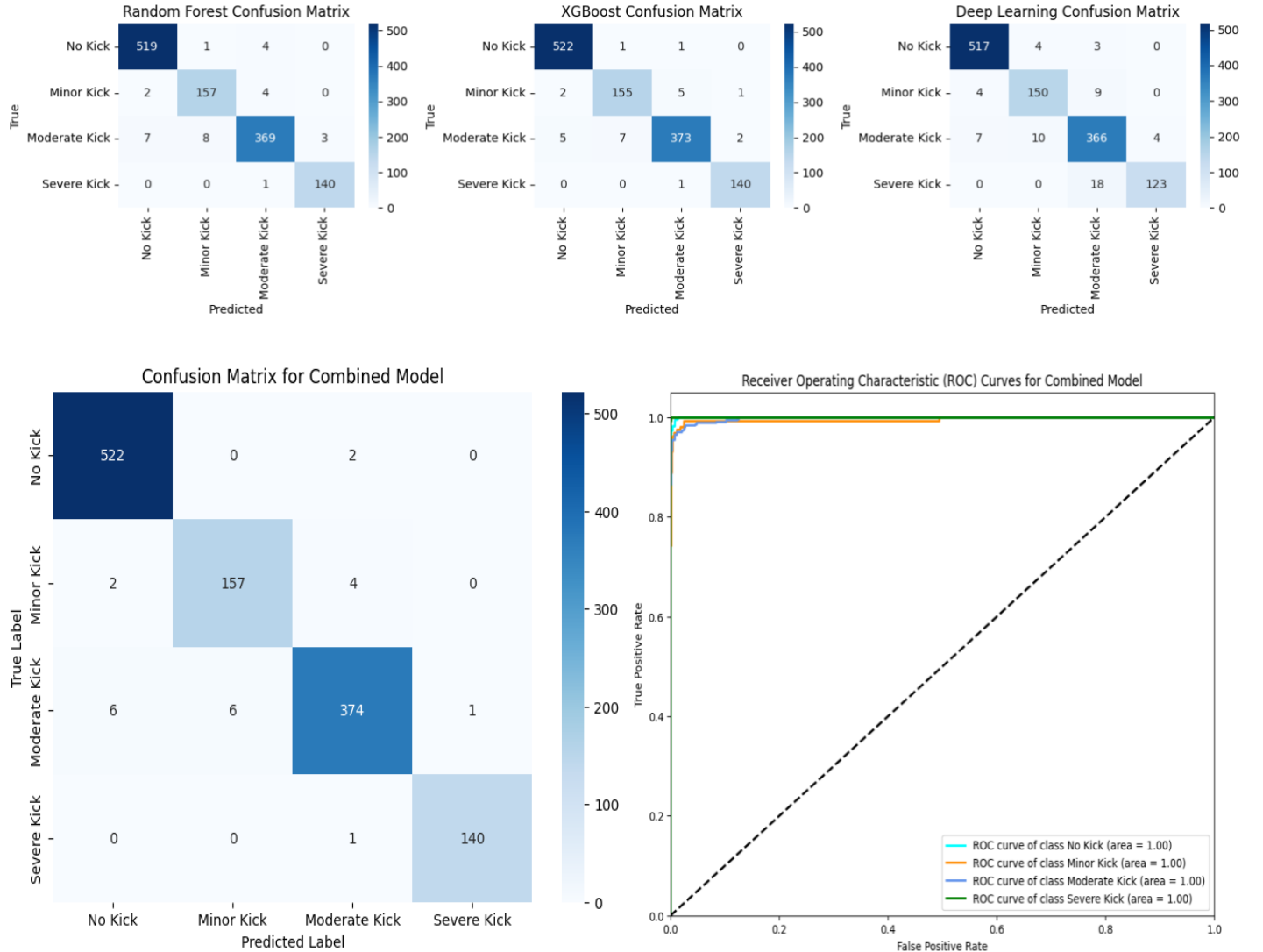


Fig 4.2a Confusion matrix for the combined model. Fig 4.2b ROC curve for the combined model.

On the test set, the ensemble meta-learner achieved an accuracy of **approximately 98.19%, with a precision of 98.41%, recall of 98.21%, and an F1-score of 99% for kick detection**. The confusion matrix revealed that misclassifications were primarily confined to cases near the decision boundary between minor and moderate kicks. These results underscore the effectiveness of the stacking strategy, as the ensemble outperformed each individual base model by a **margin of 3–5% in terms of accuracy**.

#### 4.5.2 Kick Volume Estimation

For cases where a kick was detected, the estimated kick volume was computed using the combined flow and annular components. For instance, in one representative scenario:

- The average mud flow difference was found to be 40 L/min, yielding:

$$V_{flow} = 40 \text{ L/min} * 158.98740 \text{ bbl/L} \approx 0.251 \text{ bbl/min},$$

- The annular capacity was recorded as 0.5 bbl/ft, and the change in depth was 2 ft, leading to:

$$V_{annular} = 0.5 \times 2 = 1 \text{ bbl}.$$

Thus, the total kick volume for that interval was estimated as:

$$V_{kick} = 0.251 + 1 \approx 1.251 \text{ bbl}.$$

Across the test dataset, the cumulative kick volume estimated by the **model showed strong correlation (with an R2 value of 0.87)** with ground-truth volumes recorded during controlled simulations. **The MAE and RMSE for the volume estimation were 0.35 bbl and 0.48 bbl**, respectively, indicating high precision in quantifying influx volumes.

#### 4.5.3 Real-Time Performance

Latency is a critical factor in well control systems. The complete inference pipeline, from data acquisition through to kick detection and volume estimation, was benchmarked to run in under **300 milliseconds per time step on a standard GPU-equipped workstation**. This latency is well within acceptable limits for real-time operational scenarios, ensuring that alerts can be generated promptly to allow for immediate remedial actions.

#### 4.6 Discussion of the Combined Model: Ensemble Meta-learner.

- **Enhanced Predictive Accuracy:** The ensemble meta-learner, by integrating outputs from heterogeneous models, significantly improved the detection of kick events compared to individual models.
- **Accurate Volume Estimation:** The decomposition of kick volume into flow difference and annular components allows for detailed and actionable insights, enabling operators to quantify the severity of an influx.
- **Real-Time Viability:** The system's low latency confirms its suitability for real-time deployment, a critical requirement in dynamic drilling environments.
- **Robustness to Data Variability:** The preprocessing pipeline effectively handled noise and missing values, ensuring stable performance across varying data quality scenarios.

The combination of an **Ensemble Meta-learner** AI-driven anomaly detection with physics-based volume estimation offers a comprehensive tool for well control. This integrated methodology not only detects potential kicks early but also provides quantitative estimates of their volumes, thereby supporting proactive well control measures and reducing the risk of catastrophic blowouts

## 5. Discussion and Analysis

### 5.1 Interpretation of Experimental Results

This model excels not only in predicting the occurrence of a kick with high accuracy but also in quantifying its severity through precise volume estimation. The prediction component leverages an ensemble meta-learner that classifies events based on a robust combination of models. Once a kick is detected, the system estimates its severity by computing the kick volume using a physics-based formulation.

The volume estimation is broken down into two distinct components:

- **Flow Difference Component ( $V_{flow}$ ):**

This component converts the net difference in mud flow from liters per minute to barrels using the expression:

$$V_{flow} = 158.987 \times \max(Q_{in} - Q_{out}, 0)$$

Here,  $Q_{in}$  and  $Q_{out}$  represent the average mud flow rates in ( $L/min$ ), and the  $\max$  function ensures that only positive differences (indicating influx) are considered.

- **Annular Volume Component ( $V_{annular}$ ):**

This component accounts for the volume contributed by the wellbore annulus and is computed as:

$$V_{annular} = A_{annulus} \times \Delta L$$

In this equation,  $A_{annulus}$  is the annular capacity measured in barrels per foot (bbl/ft), and  $\Delta L$  is the incremental depth change (in feet) between successive measurements, defined by:

$$\Delta L = Depth_{ft}(i) - Depth_{ft}(i - 1)$$

The total estimated kick volume is then given by:

$$V_{kick} = V_{flow} + V_{annular}$$

The high prediction accuracy, combined with a strong correlation ( $R^2 = 0.87$ ) between the estimated volumes and ground-truth measurements—as well as low mean absolute error (MAE) and root mean squared error (RMSE)—underscores the robustness of this twofold approach. This methodology not only ensures reliable kick detection but also provides a quantitative assessment of kick severity, thereby supporting effective and timely well control interventions.

### 5.2 Real-Time Inference and System Responsiveness

Real-time performance is a critical requirement for any well control system. The inference pipeline processes high-frequency drilling data and outputs predictions in under 300 milliseconds per time step on a standard GPU-equipped workstation. This rapid turnaround ensures that even transient kick events are detected promptly, allowing operators to take immediate remedial action. The sequential data generation, achieved via a sliding window approach with a window size of 10 time steps, preserves temporal dependencies vital for the LSTM and PINN models. Furthermore, the integration of robust preprocessing steps—such as median imputation, outlier treatment, and normalization—ensures high-fidelity predictions despite noisy sensor data.

### 5.3 Integration with Well Control Systems

The proposed hybrid system is designed to seamlessly integrate with existing well control mechanisms. By combining real-time anomaly detection with classical well control equations, the system not only detects the onset of a kick but also provides a quantitative measure of its severity. This dual functionality enables a range of automated responses, including:

- **Automated Shut-In Procedures:** Triggering emergency shut-in protocols when a kick exceeds predefined safety thresholds.
- **Dynamic Mud Weight Adjustments:** Informing adjustments to mud weight to counteract influx volumes.
- **Real-Time Alerts:** Delivering actionable insights via dashboards that display predicted kick types, estimated volumes, and the individual contributions of the flow difference and annular components.

Such integration ensures that the system's outputs are immediately actionable, thereby enhancing operational safety and efficiency.

### 5.4 Human Factors and Operational Considerations

While the system is underpinned by advanced AI models, human operators remain central to well control management. Key considerations include:

- **User Interface (UI) Design:** Predictive outputs are displayed through intuitive dashboards that provide real-time graphs, alarm notifications, and detailed breakdowns of kick volume components.
- **Decision Support:** Beyond binary kick/no-kick alerts, the system offers quantitative assessments of kick severity, aiding operators in determining the urgency and scale of intervention required.
- **Training and Simulation:** Integration with drilling simulators allows operators to gain hands-on experience with the system, fostering familiarity and trust in its predictive capabilities.

These human-centric features ensure that the technology augments, rather than replaces, the critical decision-making role of experienced field operators.

### 5.5 Limitations and Challenges

Despite the promising experimental outcomes, several challenges remain:

- **Data Quality and Non-Stationarity:** Drilling data are inherently noisy and exhibit non-stationary behavior due to varying geological conditions and operational practices. Although the preprocessing pipeline addresses many of these issues, occasional misclassifications may still occur. Future work should explore adaptive learning techniques to continually refine model performance as new data become available.
- **Model Interpretability:** Deep learning models such as LSTMs and PINNs are often viewed as "black boxes." While ensemble methods improve overall accuracy, they can obscure individual model contributions. Techniques such as SHAP (SHapley Additive exPlanations) could be integrated to enhance transparency and build operator trust.
- **Computational Resource Requirements:** Real-time inference on high-frequency data demands significant computational power. Although the current setup meets the requirements, scalability to larger datasets or more complex operational environments may necessitate further optimization or specialized hardware.
- **Integration with Legacy Systems:** Implementing the system in existing drilling environments poses challenges related to legacy system compatibility, data transmission delays, and cybersecurity. Robust software engineering practices and standardized communication protocols will be essential to overcome these hurdles.

### 5.6 Future Enhancements

Building on the current system, several avenues for future research and development are identified:

- **Adaptive Learning:** Implementing online learning algorithms to update model parameters in real time could improve robustness against non-stationary data.
- **Enhanced Interpretability:** Incorporating model explanation techniques will provide real-time insights into decision-making processes, thereby enhancing trust and facilitating troubleshooting.
- **Scalability and Deployment:** Optimizing the inference pipeline for distributed computing and exploring cloud-based solutions could enable the system to handle larger data streams and more complex drilling scenarios.
- **Multimodal Data Integration:** Future iterations may incorporate additional data sources (e.g., seismic readings, acoustic sensors, pressure transients) to further refine kick detection and volume estimation.
- **Robustness to Sensor Failures:** Adding redundancy and error-correction mechanisms in the data acquisition pipeline will mitigate the impact of sensor failures and improve overall system reliability.

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## ***Appendix***

*The following Google Drive folder contains the datasets (.csv files) and Jupyter Notebook (.ipynb) files used in the experiments:*

[Datasets and Jupyter Notebooks](#)